

Chapter 3: Techniques of Integrations

3.1	Integration by Parts	2
3.2	Trigonometric Integrals	8
3.3	Trigonometric Substitution	17
3.4	Rational Functions by Partial Fractions	21
3.5	Numerical Integration I: Midpoint Rule & Trapezoidal Rule	26
3.6	Numerical Integration II: Simpson's Rule	32

3.1 Integration by Parts

Reference: Stewart Calculus 8E, Section 7.1

IKIP $\rightarrow$ IL Method

Goal: To integrate product of two functions.

Every differentiation rule has a corresponding integration rule. For instance, the Substitution Rule for integration corresponds to the Chain Rule for differentiation. The rule that corresponds to the Product Rule for differentiation is called **integration by parts**.

Recall that the Product Rule states that

$$\frac{d}{dx}[u(x)v(x)] = \underbrace{u(x)v'(x)} + \underbrace{u'(x)v(x)}.$$

In the notation for indefinite integrals, we have

$$\int (\underbrace{u(x)v'(x)} + \underbrace{u'(x)v(x)}) dx = u(x)v(x).$$

We can rearrange this as

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

This formula is called the **integration-by-parts formula**.

Integration by Parts:

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

In the case of a definite integral, we have

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

Here is a general guide when applying the integration-by-parts formula to compute $\int f(x) dx$:

- (i) Identify $u(x)$ and $v'(x)$ such that $f(x) = u(x)v'(x)$.
- (ii) Choose $u(x)$ so that it is easy to integrate $u'(x)v(x)$.
- (iii) Choose $v'(x)$ so that it is easy to integrate $v'(x)$ to get $v(x)$.

increasing ease of integrating
 ↓
 Log
 Inverse Trigo
 Algebra
 Trigo
 Exponential

Example 3.1. Find $\int x \sin x dx$.

Solution. Choose $u(x) = x$, and $v'(x) = \sin x$. This implies that

$$u'(x) = 1, \quad v(x) = -\cos x.$$

By the integration-by-parts formula,

$$\begin{aligned} \int x \sin x dx &= x(-\cos x) - \int 1 \cdot (-\cos x) dx \\ &= -x \cos x + \int \cos x dx \\ &= -x \cos x + \sin x + C. \end{aligned}$$

↓
only add in at the end

□

Example 3.2. Evaluate $\int \ln x dx$.

Solution. Choose $u(x) = \ln x$, $v'(x) = 1$. This implies that

$$u'(x) = \frac{1}{x}, \quad v(x) = x.$$

By the integration-by-parts formula,

$$\begin{aligned} \int \ln x \, dx &= \overset{\text{I}}{x} \overset{\text{K}}{\ln x} - \overset{\text{J}}{\int} \overset{\text{I}}{\frac{1}{x}} \cdot \overset{\text{I}}{x} \, dx \\ &= x \ln x - \int 1 \, dx \\ &= x \ln x - x + C. \end{aligned}$$

□

Derivatives of Inverse Trigonometric Functions:

$$\begin{aligned} \frac{d}{dx}(\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\csc^{-1} x) &= -\frac{1}{x\sqrt{x^2-1}} \\ \frac{d}{dx}(\cos^{-1} x) &= -\frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\sec^{-1} x) &= \frac{1}{x\sqrt{x^2-1}} \\ \frac{d}{dx}(\tan^{-1} x) &= \frac{1}{1+x^2} & \frac{d}{dx}(\cot^{-1} x) &= -\frac{1}{1+x^2} \end{aligned}$$

→ given in exam

Example 3.3. Calculate $\int_0^1 \tan^{-1} x \, dx$.

Solution. Choose $u(x) = \tan^{-1} x$, $v'(x) = 1$. This implies that

$$u'(x) = \frac{1}{1+x^2}, \quad v(x) = x.$$

By the integration-by-parts formula,

$$\begin{aligned} \overset{\text{J}}{\int_0^1} \tan^{-1} x \, dx &= \overset{\text{I}}{x} \overset{\text{K}}{\tan^{-1} x} \Big|_0^1 - \overset{\text{J}}{\int_0^1} \overset{\text{I}}{\frac{1}{1+x^2}} \cdot \overset{\text{I}}{x} \, dx \\ &= (1 \cdot \tan^{-1} 1 - 0 \cdot \tan^{-1} 0) - \int_0^1 \frac{x}{1+x^2} \, dx \\ &= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} \, dx. \end{aligned}$$

To evaluate $\int_0^1 \frac{x}{1+x^2} dx$, we use the substitution $u = 1 + x^2$. Note that $x = 0 \implies u = 1$, and $x = 1 \implies u = 2$, and $du = 2x dx$. Thus,

$$\begin{aligned} \int_0^1 \frac{x}{1+x^2} dx &= \int_1^2 \frac{1}{2u} du \\ &= \frac{1}{2} [\ln u]_1^2 \\ &= \frac{1}{2} \ln 2. \end{aligned}$$

Hence,

$$\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \frac{\ln 2}{2}.$$

□

Note: Sometimes we may need to apply integration-by-parts **more than once** to obtain an answer.

Example 3.4. Evaluate $\int e^x \sin x dx$.

Solution. Choose $u(x) = e^x$, $v'(x) = \sin x$. This implies that

$$u'(x) = e^x, \quad v(x) = -\cos x.$$

By the integration-by-parts formula,

$$\begin{aligned} \int e^x \sin x dx &= \overset{\text{u}}{e^x} (-\overset{\text{v}}{\cos x}) - \int \overset{\text{u}'}{e^x} (-\overset{\text{v}'}{\cos x}) dx \\ &= -e^x \cos x + \int e^x \cos x dx. \end{aligned} \tag{3.1}$$

We can deal with $\int e^x \cos x dx$ by using integration by parts again.

$$\int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx.$$

Substituting this back into (3.1),

$$\begin{aligned}\int e^x \sin x \, dx &= -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx \\ 2 \int e^x \sin x \, dx &= e^x (\sin x - \cos x) \\ \int e^x \sin x \, dx &= \frac{1}{2} e^x (\sin x - \cos x) + C.\end{aligned}$$

□

Does the integration-by-parts formula still work if we were to choose $u(x) = \sin x$ and $v'(x) = e^x$?

Example 3.5. Prove the reduction formula:

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx.$$

where $n \geq 2$ is an integer.

Solution. Choose $u(x) = \sin^{n-1} x$, $v'(x) = \sin x$. This implies that

$$u'(x) = (n-1) \sin^{n-2} x \cos x, \quad v(x) = -\cos x.$$

By the integration-by-parts formula,

$$\begin{aligned}\int \sin^n x \, dx &= \overset{\text{K}}{\sin^{n-1} x} \overset{\text{I}}{(-\cos x)} - \int \overset{\text{D}}{(n-1) \sin^{n-2} x \cos x} \overset{\text{I}}{(-\cos x)} \, dx \\ &= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx.\end{aligned}$$

Since $\cos^2 x = 1 - \sin^2 x$, we have

$$\begin{aligned}\int \sin^n x \, dx &= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx \\ &= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx \\ n \int \sin^n x \, dx &= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx\end{aligned}$$

Hence,

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx.$$

□

Some of you might have learned about the LIATE rule which suggests to choose the function $u(x)$ that comes firsts in the following list:

L	Logarithmic functions
I	Inverse Trigo functions
A	Algebracu functions
T	Trigo functions
E	Exponentital functions

Beware that the rule **does not always work!**

Example 3.6. Evaluate $\int \frac{x^3 \sin(x^2)}{u \cdot v'} \, dx$.

Solution. Give it a try!

$$x^3 \cdot \sin(x^2) = \underbrace{x^2}_u \cdot \underbrace{x \sin x^2}_{v'}$$

$$u' = 2x$$

$$v = \int x \sin x^2 \, dx$$

$$= \frac{1}{2} \int 2x \sin x^2 \, dx \quad \text{use substitution}$$

$$= -\frac{1}{2} \cos x^2$$

$$\int x^3 \sin x^2 \, dx$$

$$= \int x^2 \cdot x \sin x^2 \, dx$$

$$= x^2 \left(-\frac{1}{2} \cos x^2 \right) - \int 2x \left(-\frac{1}{2} \cos x^2 \right) dx$$

$$= -\frac{x^2 \cos x^2}{2} - \int x \cos x^2 \, dx$$

$$= -\frac{x^2 \cos x^2}{2} + \frac{1}{2} \sin x^2 + C$$

use substitution $w = x^2$
 $\frac{dw}{dx} = 2x$
 $dw = 2x dx$

$$\frac{1}{2} \int 2x \cos x^2 \, dx$$

$$= \frac{1}{2} \int \cos w \, dw$$

$$= \frac{1}{2} \sin w + C$$

$$= \frac{1}{2} \sin(x^2)$$

3.2 Trigonometric Integrals

Reference: Stewart Calculus 8E, Section 7.2

Goal: Use trigonometric identities to integrate certain combinations of trigonometric functions.

→ provided for exams

Basic Idea: Reduce power

Example 3.7. Evaluate $\int \cos^3 x \, dx$.

Solution. Strategy: Reduce the powers.

Notice the power is odd. Save one cosine factor and convert the remaining $\cos^2 x$ factor to an expression involving sine using the identity:

$$\sin^2 x + \cos^2 x = 1.$$

$$\begin{aligned} \int \cos^3 x \, dx &= \int \cos x \cdot \cos^2 x \, dx \\ &= \int \cos x \cdot (1 - \sin^2 x) \, dx \\ &= \int (\cos x - \cos x \sin^2 x) \, dx \\ &= \int \cos x \, dx - \int \cos x \sin^2 x \, dx \\ &= \sin x - \int \cos x \sin^2 x \, dx \end{aligned} \tag{3.2}$$

We continue evaluating $\int \cos x \sin^2 x \, dx$ using Substitution Rule.

Set $u = \sin x$ so that $du = \cos x \, dx$. Then

$$\begin{aligned} \int \cos x \sin^2 x \, dx &= \int u^2 \, du \\ &= \frac{u^3}{3} + C \\ &= \frac{\sin^3 x}{3} + C. \end{aligned}$$

Substituting this back into (3.2),

$$\int \cos^3 x \, dx = \sin x - \frac{\sin^3 x}{3} + C.$$

□

Example 3.8. Find $\int \sin^4 x \, dx$.

Solution. Strategy: Reduce the power.

Notice the power is even. Make use of half-angle formula:

Half-angle formula:

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x), \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin x \cos x = \frac{1}{2} \sin 2x.$$

$$\begin{aligned} \int \sin^4 x \, dx &= \int (\sin^2 x)^2 \, dx \\ &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx \quad (\text{by a half-angle formula}) \\ &= \frac{1}{4} \int (1 - 2 \cos 2x + \cos^2 2x) \, dx \end{aligned}$$

Since the term $\cos^2 2x$ is a cosine with even power, we use another half-angle formula to write

$$\cos^2 2x = \frac{1}{2}(1 + \cos 4x).$$

This gives

$$\begin{aligned} \int \sin^4 x \, dx &= \frac{1}{4} \int \left(1 - 2 \cos 2x + \frac{1}{2}(1 + \cos 4x) \right) \, dx \\ &= \frac{1}{4} \int \left(\frac{3}{2} - 2 \cos 2x + \frac{1}{2} \cos 4x \right) \, dx \\ &= \frac{1}{4} \left(\frac{3}{2}x - \sin 2x + \frac{1}{8} \sin 4x \right) + C. \end{aligned}$$

□



Strategy for evaluating $\int \sin^m x \cos^n x dx$:

(a) n is **odd**. $\rightarrow 2k+1$

Save one cosine factor, and express the rest in terms of sines using

$$\cos^2 x + \sin^2 x = 1.$$

Then substitute $u = \sin x$. $du = \cos x dx$

(b) m is **odd**. $\rightarrow 2k+1$

Save one sine factor and express the rest in terms of cosines using

$$\cos^2 x + \sin^2 x = 1.$$

Then substitute $u = \cos x$. $du = -\sin x dx \Rightarrow -du = \sin x dx$

(c) Both m and n are **even**.

Use half-angle formulas.

If both m and n are odd,
you use either (a) or (b).

Example 3.9. Find $\int \sin^5 x \cos^2 x dx$.

Solution. The strategy is:

- Save one sine factor
- Convert everything else into cosine.
- Substitute $u = \cos x$

Write

$$\sin^5 x \cos^2 x = \sin^4 x \cos^2 x \sin x,$$

where the last sine factor is the one we saved.

Convert all the rest (except the one sine factor we saved) into cosines:

$$\begin{aligned}\sin^5 x \cos^2 x &= (\sin^2 x)^2 \cos^2 x \sin x \\ &= (1 - \cos^2 x)^2 \cos^2 x \sin x\end{aligned}$$

Substitute $u = \cos x$. Then $du = (-\sin x) dx \implies (-du) = \sin x dx$, and so

$$\begin{aligned}\int \sin^5 x \cos^2 x dx &= \int (1 - u^2)^2 (u^2) (-du) \\ &= \int (1 - 2u^2 + u^4)(-u^2) du \\ &= \int (-u^2 + 2u^4 - u^6) du \\ &= -\frac{u^3}{3} + \frac{2u^5}{5} - \frac{u^7}{7} + C \\ &= -\frac{1}{3} \cos^3 x + \frac{2}{5} \cos^5 x - \frac{1}{7} \cos^7 x + C.\end{aligned}$$

□

Next, we look at a similar strategy for evaluating $\int \tan^m x \sec^n x \, dx$.

Strategy for evaluating $\int \tan^m x \sec^n x \, dx$:

(a) n is **even**.

Save one $\sec^2 x$, and express the remaining factors in terms of $\tan x$ using

$$\tan^2 x + 1 = \sec^2 x.$$

Then substitute $u = \tan x$. $du = \sec^2 x \, dx$

(b) m is **odd**.

Save one $\sec x \tan x$, and express the remaining factors in terms of $\sec x$ using

$$\tan^2 x + 1 = \sec^2 x.$$

Then substitute $u = \sec x$. $du = \sec x \tan x \, dx$

Other useful formulas:

$$\int \tan x \, dx = \ln |\sec x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C.$$

doesn't cover all cases
↓
can consider integration by parts if doesn't fall into these two cases

Example 3.10. Evaluate $\int \tan^6 x \sec^4 x \, dx$.

Solution. The strategy is:

- Save one factor of $\sec^2 x$.
- Express the remaining factors in $\tan x$.
- Substitute $u = \tan x$.

Note that for the substitution $u = \tan x$, we have

$$du = \sec^2 x \, dx.$$

$$\begin{aligned}
\int \tan^6 x \sec^4 x \, dx &= \int \tan^6 x \sec^2 x \sec^2 x \, dx \\
&= \int \tan^6 (1 + \tan^2 x) \sec^2 x \, dx \\
&= \int u^6 (1 + u^2) \, du \\
&= \int u^6 + u^8 \, du \\
&= \frac{u^7}{7} + \frac{u^9}{9} + C \\
&= \frac{1}{7} \tan^7 x + \frac{1}{9} \tan^9 x + C.
\end{aligned}$$

□

Example 3.11. Find $\int \tan^5 x \sec^7 x \, dx$

Solution. The strategy is:

- Save one factor of $\sec x \tan x$.
- Express the remaining factors in terms of $\sec x$.
- Substitute $u = \sec x$.

Note that for the substitution $u = \sec x$, we have

$$du = \sec x \tan x \, dx.$$

$$\begin{aligned}
\int \tan^5 x \sec^7 x \, dx &= \int \tan^4 x \sec^6 x (\sec x \tan x) \, dx \\
&= \int (\sec^2 x - 1)^2 \sec^6 x (\sec x \tan x) \, dx \\
&= \int (u^2 - 1)^2 u^6 \, du \\
&= \int u^{10} - 2u^8 + u^6 \, du \\
&= \frac{u^{11}}{11} - 2\frac{u^9}{9} + \frac{u^7}{7} + C \\
&= \frac{1}{11} \sec^{11} x - \frac{2}{9} \sec^9 x + \frac{1}{7} \sec^7 x + C.
\end{aligned}$$

□

Example 3.12. Find $\int \tan^3 x \, dx$.

Solution. There is no $\sec x$. Try to reduce the power. We use $\tan^2 x + 1 = \sec^2 x$ to express a $\tan^2 x$ in terms of $\sec^2 x$.

$$\begin{aligned}
\int \tan^3 x \, dx &= \int \tan x \tan^2 x \, dx \\
&= \int \tan x (\sec^2 x - 1) \, dx \\
&= \int \tan x \sec^2 x \, dx - \int \tan x \, dx
\end{aligned}$$

We already know the formula $\int \tan x \, dx = \ln |\sec x| + C$.

It remains to evaluate $\int \tan x \sec^2 x \, dx$. Since the power of $\sec x$ is even, the strategy is

- Save one factor of $\sec^2 x$.
- Express the remaining factors in $\tan x$.
- Substitute $u = \tan x$.

Therefore,

$$\begin{aligned}\int \tan x \sec^2 x \, dx &= \int u \, du \\ &= \frac{u^2}{2} + C \\ &= \frac{\tan^2 x}{2} + C.\end{aligned}$$

Hence,

$$\int \tan^3 x \, dx = \frac{\tan^2 x}{2} - \ln |\sec x| + C.$$

□

To evaluate integrals

$$\int \sin mx \cos nx \, dx, \quad \int \sin mx \sin nx \, dx, \quad \int \cos mx \cos nx \, dx$$

use the corresponding **Product-to-Sum** identity:

$$(a) \quad \sin A \cos B = \frac{1}{2} (\sin(A - B) + \sin(A + B))$$

$$(b) \quad \sin A \sin B = \frac{1}{2} (\cos(A - B) - \cos(A + B))$$

$$(c) \quad \cos A \cos B = \frac{1}{2} (\cos(A - B) + \cos(A + B))$$

When simplifying, remember that:

$$\cos(-\theta) = \cos \theta, \quad \sin(-\theta) = -\sin \theta.$$

Example 3.13. Evaluate $\int \sin 4x \cos 5x \, dx$.

Solution.

$$\begin{aligned} \int \sin 4x \cos 5x \, dx &= \int \frac{1}{2} (\sin(-x) + \sin 9x) \, dx \\ &= \frac{1}{2} \int (-\sin x + \sin 9x) \, dx \\ &= \frac{1}{2} \left(\cos x - \frac{1}{9} \cos 9x \right) + C. \end{aligned}$$

□

3.3 Trigonometric Substitution

Reference: Stewart Calculus 8E, Section 7.3

Goal: To integrate via Trigonometric substitution. The integrand usually contains expressions like $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$.

coef. of x^2 may not be 1

Table of Trigonometric Substitution:

$x = g(\theta)$ $\theta = g^{-1}(x)$

Expression	Substitution	Identity involved
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ <i>$\theta = \sin^{-1}(\frac{x}{a})$</i>	$\sin^2 \theta + \cos^2 \theta = 1$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$\tan^2 \theta + 1 = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\tan^2 \theta + 1 = \sec^2 \theta$

Strategy:

We make the **inverse substitution** $x = g(\theta)$:

$$\int f(x) dx = \int f(g(\theta))g'(\theta) d\theta.$$

$u = g(x) \Rightarrow \frac{du}{dx} = g'(x) \Rightarrow du = g'(x) dx$

$\int \underbrace{f(g(x))g'(x)}_u dx = \int f(u) du$

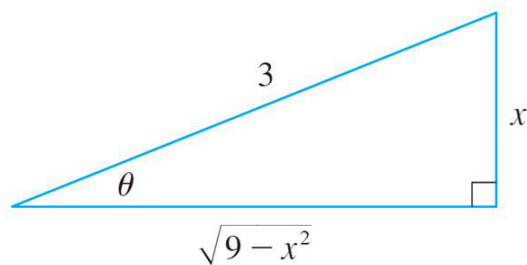
After simplifications, the right-hand side will be **expressed in x** again by substituting $\theta = g^{-1}(x)$. This is alright by **assuming g is one-to-one** (and so it would have an inverse g^{-1}). To make g one-to-one, we restrict θ to **lie in an appropriate interval** (as given in the table above).

Example 3.14. Evaluate $\int \frac{\sqrt{9 - x^2}}{x^2} dx$.

Solution. Let $x = 3 \sin \theta$. Then $dx = 3 \cos \theta d\theta$.

$$\begin{aligned}
 \int \frac{\sqrt{9-x^2}}{x^2} dx &= \int \frac{\sqrt{9-9\sin^2\theta}}{9\sin^2\theta} \cdot 3\cos\theta d\theta \\
 &= \int \frac{9\cos^2\theta}{9\sin^2\theta} d\theta \\
 &= \int \cot^2\theta d\theta \\
 &= \int \csc^2\theta - 1 d\theta \\
 &= -\cot\theta - \theta + C.
 \end{aligned}$$

$\sqrt{x^2} \neq x$ $\sqrt{(-2)^2} = 2 \neq -2$
 $\sqrt{x^2} = |x|$
 $\sqrt{\cos^2\theta} = |\cos\theta|$ $\therefore$ restriction on θ
 $= \cos\theta$ for this q^n
 $\downarrow$
 $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$



We need to convert the expressions back to that in x . Clearly, we can write

$$\theta = \sin^{-1}\left(\frac{x}{3}\right).$$

Based on the right-angle triangle above, we have

$$\tan\theta = \frac{x}{\sqrt{9-x^2}},$$

and so

$$\cot\theta = \frac{\sqrt{9-x^2}}{x}.$$

Although the right-angled triangle is only valid for $0 < \theta < \frac{\pi}{2}$, the expression we get is valid for all θ .

Hence,

$$\int \frac{\sqrt{9-x^2}}{x^2} dx = -\frac{\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C.$$

□

Example 3.15. Find $\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx$.

Solution. First note that $(4x^2 + 9)^{3/2} = \sqrt{4x^2 + 9}^3$, so trigonometric substitution is appropriate.

However, $\sqrt{4x^2 + 9}$ is not in the form $\sqrt{a^2 + x^2}$. But we can get around this by doing the following:

- Make a preliminary substitution $u = 2x$, giving

$$\sqrt{4x^2 + 9} = \sqrt{u^2 + 9}.$$

- Substitute $u = 3 \tan \theta$, where $-\pi/2 < \theta < \pi/2$, so that

$$\sqrt{4x^2 + 9} = \sqrt{u^2 + 9} = \sqrt{9 \tan^2 \theta + 9} = 3\sqrt{\tan^2 \theta + 1} = 3 \sec \theta.$$

The above two steps can be combined by taking the substitution:

$$x = \frac{3}{2} \tan \theta.$$

Then

- $dx = \frac{3}{2} \sec^2 \theta d\theta$
- $x = 0 \implies \theta = 0$
- $x = \frac{3\sqrt{3}}{2} \implies \theta = \frac{\pi}{3}$.
- $4x^2 + 9 = 4(3/2 \tan \theta)^2 + 9 = 9(\tan^2 \theta + 1) = 9 \sec^2 \theta$

So

$$\begin{aligned} \int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx &= \int_0^{\pi/3} \frac{(\frac{3}{2} \tan \theta)^3}{(9 \sec^2 \theta)^{3/2}} \cdot \frac{3}{2} \sec^2 \theta d\theta \\ &= \frac{3}{16} \int_0^{\pi/3} \frac{\tan^3 \theta}{\sec \theta} d\theta \\ &= \frac{3}{16} \int_0^{\pi/3} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta \\ &= \frac{3}{16} \int_0^{\pi/3} \frac{1 - \cos^2 \theta}{\cos^2 \theta} \sin \theta d\theta \end{aligned}$$

Substitute $u = \cos \theta$. Then $du = (-\sin \theta)d\theta \implies (-du) = \sin \theta d\theta$. Also, $\theta = 0 \implies u = \cos 0 = 1$, and $\theta = \pi/3 \implies u = \cos \pi/3 = 1/2$. So

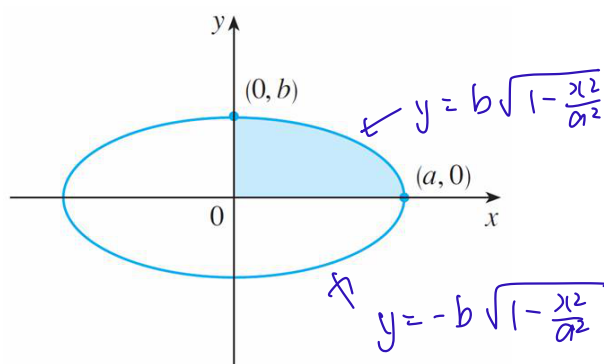
$$\begin{aligned} \int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2+9)^{3/2}} dx &= \frac{3}{16} \int_1^{1/2} \frac{1-u^2}{u^2} (-du) \\ &= \frac{3}{16} \int_1^{1/2} (1-u^{-2}) du \\ &= \frac{3}{16} [u + u^{-1}]_1^{1/2} \\ &= \frac{3}{32}. \end{aligned}$$

□

Example 3.16. Find the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Solution. Because the ellipse is symmetric with respect to both axes, the area A of the region enclosed by the ellipse is **four times the area in the first quadrant**



Give it a try!

Assume a is +ve

$$\begin{aligned} \text{Area} &= \int_{-a}^a \left(b\sqrt{1-\frac{x^2}{a^2}} - (-b\sqrt{1-\frac{x^2}{a^2}}) \right) dx \\ &= 2 \int_{-a}^a \left(b\sqrt{1-\frac{x^2}{a^2}} \right) dx \\ &= 4 \int_0^a \left(b\sqrt{1-\frac{x^2}{a^2}} \right) dx \\ &= 4 \int_0^a \left(\frac{b}{a} \sqrt{a^2-x^2} \right) dx \end{aligned}$$

Let $x = a \sin \theta$
 $dx = a \cos \theta d\theta$
 $x=0 \Rightarrow a \sin \theta = 0 \Rightarrow \theta = 0$
 $x=a \Rightarrow a \sin \theta = a \Rightarrow \theta = \frac{\pi}{2}$

$$\begin{aligned} \text{Area} &= \frac{4b}{a} \int_0^{\frac{\pi}{2}} \sqrt{a^2 - a^2 \sin^2 \theta} (a \cos \theta d\theta) \\ &= 4ba \int_0^{\frac{\pi}{2}} \sqrt{1 - \sin^2 \theta} (\cos \theta) d\theta \\ &= 4ba \int_0^{\frac{\pi}{2}} (\cos \theta \cdot \cos \theta) d\theta \end{aligned}$$

$$\begin{aligned} &= 4ba \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 2\theta) d\theta \\ &= 2ba \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\ &= 2ba \left(\frac{\pi}{2} \right) \\ &= \pi ab \end{aligned}$$

3.4 Rational Functions by Partial Fractions

Reference: Stewart Calculus 8E, Section 7.4

Goal: To integrate any rational function (a ratio of polynomials) by expressing it as a sum of simpler functions, called partial fractions.

A rational function of x is a function of the form $\frac{P(x)}{Q(x)}$ where both $P(x)$ and $Q(x)$ are polynomials.

This process of taking a rational function and decomposing it into simpler rational expressions that we can add or subtract to get the original rational expression is called **partial fraction decomposition**. For example,

$$\frac{x+5}{x^2+x-2} = \frac{2}{x-1} - \frac{1}{x+2}$$

is a partial fraction decomposition.

Partial Fraction Decomposition:

- Start with a rational function $\frac{P(x)}{Q(x)}$, where the ↗ denominator higher power **degree of P is strictly less than the degree of Q** , in which case we say that the rational function is **proper**. This is important, otherwise we will first need to do a long division to expand the function.
- We factor the denominator $Q(x)$ as completely as possible. For our purposes, $Q(x)$ only contains linear or quadratic factors. For each factor in the denominator we can use the following table to determine the term(s) we pick up in the partial fraction decomposition.

Factor in $Q(x)$	Term in partial fraction decomposition
$(ax+b)^r$	$\frac{A_1}{(ax+b)} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_r}{(ax+b)^r}, \quad r = 1, 2, 3, \dots$
$(ax^2+bx+c)^r$	$\frac{A_1x+B_1}{(ax^2+bx+c)} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \cdots + \frac{A_rx+B_r}{(ax^2+bx+c)^r}, \quad r = 1, 2, 3, \dots$

↓
cannot be further factored
↳ no real roots $\Rightarrow b^2 - 4ac < 0$

Example 3.17. Write out the form of the partial fraction decomposition of the function

$$\frac{x^3 + x^2 + 1}{x(x-1)(x^2 + x + 1)(x^2 + 1)^3}.$$

Solution. The denominator is already given as the product of its factors. We have

$$\begin{aligned} & \frac{x^3 + x^2 + 1}{x(x-1)(x^2 + x + 1)(x^2 + 1)^3} \\ &= \frac{A}{x} + \frac{B}{x-1} + \frac{Cx + D}{x^2 + x + 1} + \frac{Ex + F}{x^2 + 1} + \frac{Gx + H}{(x^2 + 1)^2} + \frac{Ix + J}{(x^2 + 1)^3}. \end{aligned}$$

□

Example 3.18. Evaluate $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$

Solution. We can apply the partial fraction decomposition since the degree of the numerator is strictly less than the degree of the denominator.

We factor the denominator as

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(2x - 1)(x + 2).$$

All the factors are linear and occur exactly once. So the partial fraction decomposition has the form

$$\frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}.$$

To obtain the constants A , B , C , we multiply the equation by the denominator $x(2x - 1)(x + 2)$:

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1).$$

Expanding the right-hand side, we have

$$x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x + (-2A).$$

Comparing the coefficients of x , we have

$$2A + B + 2C = 1$$

$$3A + 2B - C = 2$$

$$2A = 1.$$

Solving, we get

$$A = \frac{1}{2}, \quad B = \frac{1}{5}, \quad C = -\frac{1}{10}.$$

Thus,

$$\begin{aligned} \int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx &= \int \left(\frac{1}{2x} + \frac{1}{5(2x-1)} - \frac{1}{10(x+2)} \right) dx \\ &= \frac{1}{2} \ln|x| + \frac{1}{5} \frac{\ln|2x-1|}{2} - \frac{1}{10} \ln|x+2| + C \\ &= \frac{1}{2} \ln|x| + \frac{1}{10} \ln|2x-1| - \frac{1}{10} \ln|x+2| + C. \end{aligned}$$

□

Example 3.19. Find $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx$.

Solution. The integrand is not *proper* i.e the degree of the numerator is not less than that of the denominator. So the first step is to apply long division:

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \frac{4x}{x^3 - x^2 - x + 1}.$$

$$\begin{array}{r} x+1 \\ x^3-x^2-x+1 \overline{) x^4 } \\ \underline{-(x^4-x^3-x^2+x)} \\ x^3-x^2-2x+1 \\ \underline{-(x^3-x^2-x+1)} \\ 4x \end{array}$$

The second step is to factor the denominator $Q(x) = x^3 - x^2 - x + 1$. Notice $Q(1) = 0$, so $x - 1$ is factor of $Q(x)$. Factoring out $x - 1$ from $Q(x)$, we have

$$x^3 - x^2 - x + 1 = (x - 1)(x^2 - 1) = (x - 1)(x - 1)(x + 1) = (x - 1)^2(x + 1).$$

The partial fraction decomposition of $\frac{4x}{x^3 - x^2 - x + 1}$ is

$$\frac{4x}{x^3 - x^2 - x + 1} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{x + 1}.$$

Multiplying both sides by $(x - 1)^2(x + 1)$, we have

$$\begin{aligned} 4x &= A(x - 1)(x + 1) + B(x + 1) + C(x - 1)^2 \\ 4x &= (A + C)x^2 + (B - 2C)x + (-A + B + C). \end{aligned}$$

Comparing coefficients, we deduce that

$$\begin{aligned} A + C &= 0 \\ B - 2C &= 4 \\ -A + B + C &= 0. \end{aligned}$$

Solving, we obtain

$$A = 1, \quad B = 2, \quad C = -1.$$

Thus

$$\begin{aligned} \int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx &= \int \left(x + 1 + \frac{1}{x - 1} + \frac{2}{(x - 1)^2} - \frac{1}{x + 1} \right) dx \\ &= \frac{x^2}{2} + x + \ln|x - 1| - \frac{2}{x - 1} - \ln|x + 1| + C \\ &= \frac{x^2}{2} + x - \frac{2}{x - 1} + \ln \left| \frac{x - 1}{x + 1} \right| + C. \end{aligned}$$

□

Example 3.20. Evaluate $\int \frac{1-x+2x^2-x^3}{x(x^2+1)^2} dx$

Solution. Give it a try!

$$\frac{1-x+2x^2-x^3}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

$$1-x+2x^2-x^3 = A(x^2+1)^2 + (Bx+C)x(x^2+1) + (Dx+E)x$$

$$= Ax^4 + 2Ax^2 + A + Bx^4 + Bx^2 + Cx^3 + Cx + Dx^2 + Ex$$

$$= (A+B)x^4 + Cx^3 + (2A+B+D)x^2 + (C+E)x + A$$

$$\begin{aligned} \therefore A &= 1 \\ A+B &= 0 \\ \therefore B &= -1 \\ \therefore C &= -1 \\ 2A+B+D &= 2 \\ \therefore D &= 2-2(1)-(-1) \\ &= 1 \\ C+E &= -1 \\ \therefore E &= -1-(-1) \\ &= 0 \end{aligned}$$

$$\therefore \frac{1-x+2x^2-x^3}{x(x^2+1)^2} = \frac{1}{x} - \frac{x+1}{x^2+1} + \frac{x}{(x^2+1)^2}$$

$$\int \frac{1}{x} dx = \ln|x|$$

can use $\frac{f'(x)}{f(x)}$ formula!

$$\int \frac{x+1}{x^2+1} dx = \int \frac{x}{x^2+1} dx + \int \frac{1}{x^2+1} dx$$

$\downarrow u=x^2+1$

$$= \frac{1}{2} \ln|x^2+1| + \tan^{-1}x$$

$$\int \frac{x}{(x^2+1)^2} dx = \int \frac{1}{u^2} du = \frac{1}{2} \left(\frac{u^{-1}}{-1} \right)$$

$u=x^2+1$

$$= -\frac{1}{2u}$$

$$= -\frac{1}{2(x^2+1)}$$

□

Example 3.21. Evaluate $\int \frac{\sqrt{x+4}}{x} dx$.

Solution. Give it a try!

$$\text{Let } u = \sqrt{x+4} \Rightarrow u^2 = x+4 \Rightarrow x = u^2 - 4$$

$\downarrow$

$$2u \frac{du}{dx} = 1$$

$$dx = 2u du$$

$$\int \frac{u}{u^2-4} (2u du) = \int \frac{2u^2}{u^2-4} du$$

$$\frac{u^2-4}{u^2-4} \frac{2u^2}{u^2-4} = \frac{2u^2}{u^2-4}$$

$$\therefore \frac{2u^2}{u^2-4} = 2 + \frac{8}{u^2-4}$$

$$\therefore \int \frac{2u^2}{u^2-4} du = \int 2 + \frac{8}{u^2-4} du$$

$$= \int 2 + \frac{8}{(u+2)(u-2)} du$$

verify myself

$$= \int 2 + \frac{2}{u-2} - \frac{2}{u+2} du$$

$$= 2u + 2\ln|u-2| - 2\ln|u+2| + C$$

$$= 2\sqrt{x+4} + 2\ln\left|\frac{u-2}{u+2}\right| + C = 2\sqrt{x+4} + 2\ln\left|\frac{\sqrt{x+4}-2}{\sqrt{x+4}+2}\right| + C$$

□

3.5 Numerical Integration I: Midpoint Rule & Trapezoidal Rule

Reference: Stewart Calculus 8E, Section 7.7.

Goal: There are situations in which it is impossible to find the exact value of a definite integral. In such cases, we can approximate the values of definite integrals using the Riemann sums. We will focus on the left/right endpoint approximation, the Midpoint rule and the Trapezoidal rule.

Recall that the definite integral over an interval $[a, b]$ is defined as the limit of Riemann sums: If we divide $[a, b]$ into n subintervals of equal length $\Delta x = \frac{b-a}{n}$, then

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i^*) \Delta x,$$

without the limits

where x_i^* is any point in the i th subinterval $[x_{i-1}, x_i]$.

Thus, different choices of x_i^* give different approximations to the definite integral. In particular, choice of x_i^* determines the kind of approximations we have

- If we choose $x_i^* = x_{i-1}$ to be the left endpoint of every subinterval, then

$$\int_a^b f(x) dx \approx L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x,$$

where L_n is called the **left endpoint approximation**.

- If we choose $x_i^* = x_i$ to be the right endpoint of every subinterval, then

$$\int_a^b f(x) dx \approx R_n = \sum_{i=1}^n f(x_i) \Delta x,$$

where R_n is called the **right endpoint approximation**.

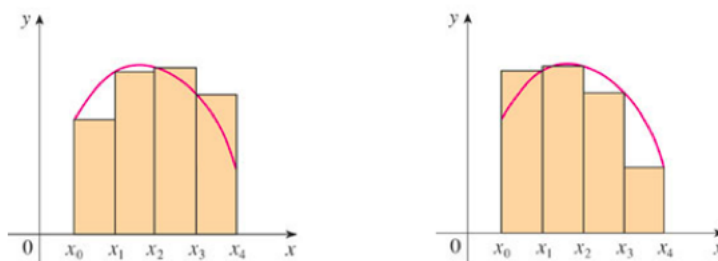


Figure 1: (Left) Left endpoint approximation. (Right) Right endpoint approximation.

If we choose $x_i^* = \frac{1}{2}(x_i + x_{i-1})$ to be the midpoint of every subinterval, then we will get the **midpoint approximation**:

averaging domain
Midpoint Rule $\rightarrow$ sum like average...

$$\int_a^b f(x) dx \approx M_n = \Delta x (f(\bar{x}_1) + \underbrace{f(\bar{x}_2) + \cdots + f(\bar{x}_n)}_{\sum_{i=1}^n f(\bar{x}_i)})$$

where

$$\Delta x = \frac{b - a}{n}$$

and

$$\bar{x}_i = \frac{1}{2}(x_i + x_{i-1}) = \text{midpoint of } [x_{i-1}, x_i].$$

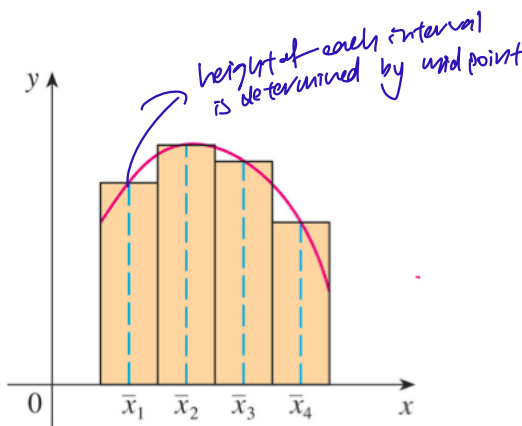


Figure 2: Midpoint rule.

Another approximation, called the Trapezoidal Rule, results from averaging the left endpoint and right endpoint approximations:

Trapezoidal Rule

$$\begin{aligned}
 \int_a^b f(x) dx &\approx T_n = \frac{1}{2}(L_n + R_n) \\
 &= \frac{1}{2} \left(\sum_{i=1}^n f(x_{i-1}) \Delta x + \sum_{i=1}^n f(x_i) \Delta x \right) \\
 &= \frac{\Delta x}{2} \left(\sum_{i=1}^n f(x_{i-1}) + f(x_i) \right) \rightarrow \text{area of trapezium} \\
 &= \frac{\Delta x}{2} ((f(x_0) + f(x_1)) + (f(x_1) + f(x_2)) + \cdots + (f(x_{n-1}) + f(x_n))) \\
 &= \frac{\Delta x}{2} (f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n))
 \end{aligned}$$

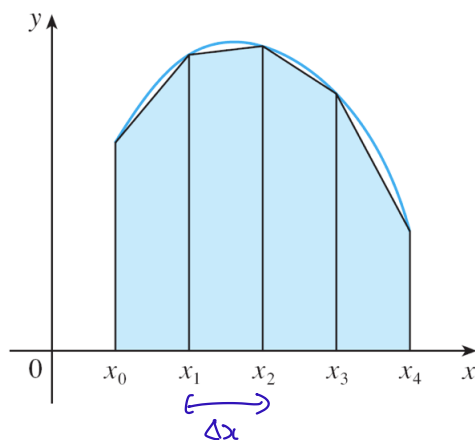
where

$$\Delta x = \frac{b-a}{n}$$

and

$$x_i = a + i\Delta x.$$

The reason for the name “Trapezoidal Rule” can be seen from the following figure:



Example 3.22. Use (a) the Trapezoidal Rule and (b) the Midpoint Rule with $n = 5$ to approximate the integral $\int_1^2 \frac{1}{x} dx$.

Solution. With $n = 5$, $a = 1$, $b = 2$, we have

$$\Delta x = \frac{b - a}{n} = \frac{2 - 1}{5} = 0.2.$$

The five subintervals are given by the points:

$$x_0 = 1, x_1 = 1.2, x_2 = 1.4, x_3 = 1.6, x_4 = 1.8, x_5 = 2.0.$$

(a) The Trapezoidal Rule gives

$$\begin{aligned} \int_1^2 \frac{1}{x} dx &\approx \frac{0.2}{2} (f(1) + 2f(1.2) + 2f(1.4) + 2f(1.6) + 2f(1.8) + f(2)) \\ &= 0.1 \left(\frac{1}{1} + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2} \right) \\ &\approx 0.695635. \end{aligned}$$

(b) The midpoints of the 5 subintervals are

$$1.1, 1.3, 1.5, 1.7, 1.9.$$

The Midpoint Rule gives

$$\begin{aligned} \int_1^2 \frac{1}{x} dx &\approx 0.2 (f(1.1) + f(1.3) + f(1.5) + f(1.7) + f(1.9)) \\ &= 0.2 \left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9} \right) \\ &\approx 0.691908. \end{aligned}$$

□

NOTE. In the example above, if we compute the integral explicitly, then we can see how accurate the approximations given by the Trapezoidal Rule and the Midpoint Rule. By the Fundamental Theorem of Calculus,

$$\int_1^2 \frac{1}{x} dx = [\ln x]_1^2 = \ln 2 \approx 0.693147.$$

Define the **error** in using an approximation to be the amount that needs to be added to the approximation to make it exact. In general, define

$$\text{error} = \text{true value} - \text{approximation}$$

$$\text{Error} > 0 \rightarrow \text{underestimation}$$

$$\text{Error} < 0 \rightarrow \text{overestimation}$$

Approximation may overestimate or underestimate the true value.
Error bounds give an upper bound on the (absolute) error.

- $E_T = \int_a^b f(x) dx - T_n$ to be the error in using the Trapezoidal Rule with n subintervals.
- $E_M = \int_a^b f(x) dx - M_n$ to be the error in using the Midpoint Rule with n subintervals.
- $E_L = \int_a^b f(x) dx - L_n$ to be the error in using the Left Endpoints approximations with n subintervals.
- $E_R = \int_a^b f(x) dx - R_n$ to be the error in using the Right Endpoints approximations with n subintervals.

The following tables show the result of calculations in approximating $\int_1^2 \frac{1}{x} dx$ for $n = 5$, $n = 10$, $n = 20$ using different methods:

n	L_n	R_n	T_n	M_n
5	0.745635	0.645635	0.695635	0.691908
10	0.718771	0.668771	0.693771	0.692835
20	0.705803	0.680803	0.693303	0.693069

n	E_L	E_R	E_T	E_M
5	-0.052488	0.047512	-0.002488	0.001239
10	-0.025624	0.024376	-0.000624	0.000312
20	-0.012656	0.012344	-0.000156	0.000078

We can make several observations from these tables:

- (1) We get more accurate results when we increase the value of n .
- (2) The errors in the left and right endpoint approximations are opposite in sign, and appear to decrease by a factor of 2 when we double the value of n .
- (3) The Trapezoidal and Midpoint Rules are much more accurate than the endpoint approximations.
- (4) The errors in the Trapezoidal and Midpoint Rules are opposite in sign and appear to decrease by a factor of 4 when we double the value of n .
- (5) The size of error in the Midpoint Rule is about half the size of error in the Trapezoidal Rule.

overestimate
underestimate
for this particular integral $\rightarrow$ it varies

The following theorem gives an upper bound on the error in Trapezoidal and Midpoint Rules. The proof is beyond the scope of our syllabus.



Theorem 1. (Error Bounds for Trapezoidal and Midpoint Rule)

Suppose $|f''(x)| \leq K$ for $a \leq x \leq b$. Then

$$|E_T| \leq \frac{K(b-a)^3}{12n^2}, \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}.$$

Example 3.23. Find an n such that the Midpoint Rule approximation M_n approximates $\int_1^4 \sqrt{x} dx$ with an error of at most 0.0001.
Goal: Find n s.t. $|E_M| \leq 0.0001$
 $\Rightarrow |E_M| \leq \frac{K(b-a)^3}{24n^2} \leq 0.0001$

Solution. Let $f(x) = \sqrt{x}$. To apply the Error Bound, we must find a number K such that $|f''(x)| \leq K$ for all $x \in [1, 4]$.

We have $f'(x) = \frac{1}{2}x^{-1/2}$, and so

$$f''(x) = -\frac{1}{4}x^{-3/2}$$

$$|f''(x)| = \frac{1}{4}x^{-3/2}$$

Note that $|f''(x)|$ is decreasing on $[1, 4]$, so its maximum occurs at $x = 1$. Thus, we may take $K = |f''(1)| = \frac{1}{4}$:

$$|f''(x)| \leq K = \frac{1}{4}, \quad \text{for all } x \in [1, 4].$$

By the Error Bound,

$$|E_M| \leq \frac{K(b-a)^3}{24n^2} = \frac{27}{96n^2} \leq 0.0001.$$

Handwritten notes: Arrows point from '4' and '1' in the numerator to 'from the integral'. An arrow points from '24' in the denominator to 'from the integral'.

Thus, the error is at most 0.0001 if

$$n^2 \geq \frac{27 \times 10^4}{96} = 2812.5.$$

$$n \geq \sqrt{2812.5} = 53.033.$$

So we can choose $n = 54$ (actually, any integer greater than 53.033 will work). $\square$

3.6 Numerical Integration II: Simpson's Rule

Reference: Stewart Calculus 8E, Section 7.7.

Goal: We exploit the idea of averaging two approximations. Using a weighted average of the Trapezoidal and Midpoint Rules, we derive a more accurate approximation called Simpson's Rule.

From our previous study, the Trapezoidal Rule approximation T_n , which is the average of the left and right endpoint approximations, seems to be more accurate than either L_n or R_n alone.

This suggests that the average of T_n and M_n may be more accurate than either T_n or M_n alone.

Simpson's Rule S_n exploits this idea, but it takes into account that M_n is roughly twice as accurate as T_n . Therefore, Simpson's rule uses a weighted average, which is more accurate than the ordinary average. Assume n is even, we define

$$S_n = \frac{1}{3}T_{n/2} + \frac{2}{3}M_{n/2}, \quad n \text{ is even.}$$

Suppose we want to approximate $\int_a^b f(x) dx$.

To derive a formula for S_n , let $\Delta x = \frac{b-a}{n}$, and divide the interval $[a, b]$ into $n/2$ subintervals:

$$[x_0, x_2], [x_2, x_4], \dots, [x_{n-2}, x_n].$$

- The endpoints of these subintervals $x_0, x_2, \dots, x_{n-2}, x_n$ are used to compute $T_{n/2}$:

$$\begin{aligned}
 T_{n/2} &= \frac{1}{2} \frac{b-a}{n/2} (f(x_0) + 2f(x_2) + 2f(x_4) + \dots + 2f(x_{n-2}) + f(x_n)) \\
 &= \Delta x (f(x_0) + 2f(x_2) + 2f(x_4) + \dots + 2f(x_{n-2}) + f(x_n)).
 \end{aligned}$$

- The midpoints of the subintervals $x_1, x_3, \dots, x_{n-3}, x_{n-1}$ are used to compute $M_{n/2}$:

$$\begin{aligned}
 M_{n/2} &= \left(\frac{b-a}{n/2} \right) (f(x_1) + f(x_3) + \dots + f(x_{n-3}) + f(x_{n-1})) \\
 &= \Delta x (2f(x_1) + 2f(x_3) + \dots + 2f(x_{n-3}) + 2f(x_{n-1}))
 \end{aligned}$$

(Handwritten note: A blue circle around $\frac{b-a}{n/2}$ with an arrow pointing to $2\Delta x$)

Therefore,

$$\begin{aligned}
 S_n &= \frac{1}{3}T_{n/2} + \frac{2}{3}M_{n/2} \\
 &= \frac{1}{3}\Delta x (f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)).
 \end{aligned}$$

Simpson's Rule

Suppose n is even.

$$\begin{aligned}
 \int_a^b f(x) dx &\approx S_n = \frac{1}{3} T_{n/2} + \frac{2}{3} M_{n/2} \\
 &= \frac{1}{3} \Delta x (f(x_0) + 4f(x_1) + 2f(x_2) + \dots \\
 &\quad + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n))
 \end{aligned}$$

where

$$\Delta x = \frac{b-a}{n}.$$

Example

if $n=2$,

$$S_2 = \frac{\Delta x}{3} (f(x_0) + 4f(x_1) + f(x_2))$$

if $n=4$,

$$S_4 = \frac{\Delta x}{3} (f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4))$$

if $n=6$,

$$S_6 = \frac{\Delta x}{3} (f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6))$$

Theorem 2. (Error Bound for Simpson's Rule)

Suppose that $|f^{(4)}(x)| \leq K$ for all $a \leq x \leq b$. If E_S is the error involved in using the Simpson's Rule, then

$$|E_S| \leq \frac{K(b-a)^5}{180n^4}.$$

Example 3.24. Use Simpson's Rule with $n = 8$ to approximate $\int_2^4 \sqrt{1+x^3} dx$.

Solution. We have $\Delta x = \frac{b-a}{n} = \frac{4-2}{8} = 0.25$, and

$$x_0 = 2, x_1 = 2.25, x_2 = 2.5, x_3 = 2.75, x_4 = 3, x_5 = 3.25, x_6 = 3.5, x_7 = 3.75, x_8 = 4.$$

Let $f(x) = \sqrt{1+x^3}$. Thus,

$$\begin{aligned} S_8 &= \frac{1}{3}(0.25)(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) \\ &\quad + 2f(x_6) + 4f(x_7) + f(x_8)) \\ &= \frac{1}{12}(\sqrt{1+2^3} + 4\sqrt{1+2.25^3} + 2\sqrt{1+2.5^3} + 4\sqrt{1+2.75^3} \\ &\quad + 2\sqrt{1+3^3} + 4\sqrt{1+3.25^3} + 2\sqrt{1+3.5^3} + 4\sqrt{1+3.75^3} + \sqrt{1+4^3}) \\ &\approx \frac{1}{12}(3 + 4(3.52003) + 2(4.07738) + 4(4.66871) + 2(5.2915) \\ &\quad + 4(5.94375) + 2(6.62382) + 4(7.33037) + 8.06226) \\ &\approx 10.74159. \end{aligned}$$

□

Example 3.25. Calculate S_8 for $\int_1^3 \frac{1}{x} dx$. Then

Error bounds for Simpson Rule depends on the 4th derivative of the function!

(i) Find a bound for the error in S_8 .

(ii) Find an n such that S_n has error of at most 10^{-6} .

Solution. We have $\Delta x = \frac{b-a}{n} = \frac{3-1}{8} = 0.25$, and

$$x_0 = 1, x_1 = 1.25, x_2 = 1.5, x_3 = 1.75, x_4 = 2, x_5 = 2.25, x_6 = 2.5, x_7 = 2.75, x_8 = 3.$$

Therefore,

$$\begin{aligned} S_8 &= \frac{1}{3}(0.25) \left(\frac{1}{1} + \frac{4}{1.25} + \frac{2}{1.5} + \frac{4}{1.75} + \frac{2}{2} + \frac{4}{2.25} + \frac{2}{2.5} + \frac{4}{2.75} + \frac{1}{3} \right) \\ &\approx 1.09873. \end{aligned}$$

Let $f(x) = \frac{1}{x}$.

- (i) The fourth derivative $f^{(4)}(x) = 24x^{-5}$ is decreasing, so its maximum on $[1, 3]$ is $f^{(4)}(1) = 24$. Therefore, we may set $K = 24$, and by using the Error Bound for Simpson's Rule, we have

$$|E_S| \leq \frac{24(3-1)^5}{180n^4} = \frac{64}{15n^4} = \frac{64}{15 \cdot 8^4} \approx 0.001.$$

- (ii) To ensure that the error is at most 10^{-6} , we choose n so that

$$|E_S| \leq \frac{64}{15n^4} \leq 10^{-6}.$$

This gives

$$\begin{aligned} n^4 &\geq 10^6 \left(\frac{64}{15} \right) \\ n &\geq \left(10^6 \left(\frac{64}{15} \right) \right)^{1/4} \approx 45.45. \end{aligned}$$

Thus, we may take $n = 46$.

□

 Note: Must choose n to be **even** for Simpson's Rule.